



## AgroLnk — Complete Production Demo & Testing Manual

### A Step-by-Step Interactive Guide for QA Testers & Stakeholder Demonstrations

#### Executive Overview

AgroLnk is an end-to-end B2B Agri-Trade Platform integrating **Spot Commerce, Live Reverse-English Auctions, 100% Escrow & 0.50% Commission, Multi-Axle Logistics, WDRA e-NWR Warehouses, 30-Day Working Capital Financing, Assayer Quality Inspections, and Anti-Circumvention Privacy Chat.**

This manual guides you step-by-step through every user role and business flow with exact account credentials, navigation paths, actions to perform, and expected results.

## 1. Test Accounts & Role Access Matrix

| USER ROLE          | PERSONA NAME           | LOGIN EMAIL             | PASSWORD             | PRIMARY PORTAL ROUTE    | KEY TESTING RESPONSIBILITIES                                                         |
|--------------------|------------------------|-------------------------|----------------------|-------------------------|--------------------------------------------------------------------------------------|
| Platform Admin     | Platform Admin         | admin@agroInk.com       | admin123<br>(or any) | #/admin-dashboard       | 1-Click KYC Approvals, Escrow Ledger Auditing, Assayer Dispatch, Dispute Arbitration |
| Verified Farmer    | Veerappan / Sakthi Vel | farmer@agroInk.com      | farmer123            | #/farmer-dashboard      | Spot Lot Creation, Live Auctions, e-NWR Warehouse Deposits, 75% LTV Loans            |
| Wholesale Buyer    | Maran S / Global Agro  | buyer@agroInk.com       | buyer123             | #/buyer-dashboard       | Direct Spot Buying, 100% Escrow Funding, Live Bidding, Pre-Dispatch Assayer Requests |
| Transporter        | Apex Fleet Logistics   | transporter@agroInk.com | pass123              | #/transporter-dashboard | Multi-Axle Fleet Allocation, Corridor Checkpoint Updates, 6-Digit Delivery OTP       |
| Financier / NBFC   | Samunnati Capital      | financier@agroInk.com   | pass123              | #/financier-dashboard   | 30-Day Credit Underwriting, Collateral Vault Liens, 1-Click Capital Disbursement     |
| Warehouse Operator | Salem Agri Cold Hub    | warehouse@agroInk.com   | pass123              | #/warehouse-dashboard   | 5,000 MT Storage Auditing, Multi-Chamber Temperature Zones, e-NWR Digital Titles     |

## 2. Step-by-Step Assistant Testing Scenarios

### SCENARIO 1: Identity KYC Gate & 1-Click Admin Approval

**Goal:** Test that unverified accounts are prevented from posting trades until Admin verifies their identity documents.

#### Step 1.1 — Login as Farmer & Attempt Listing:

**Role:** Farmer ( farmer@agroInk.com )

**Navigation:** Open sidebar ➡ Click **Post Produce** ( #/farmer-create-listing ).

**Action:** If unverified, observe the **Verification Required Modal** blocking unauthorized submission.

**Action:** Click **Submit KYC Documents** and upload an Aadhaar / GST document (Image or PDF).

#### Step 1.2 — Switch to Platform Admin:

**Role:** Admin ( `admin@agroInk.com` ).

**Navigation:** Go to **Admin Dashboard** → **KYC Verification Queue** ( `#/admin-verification` ).

**Action:** Locate the farmer in the pending queue.

**Action:** Click the  **View Document** button to preview the uploaded PDF/Image in the modal.

**Action:** Click the green  **Approve** button.

### Step 1.3 — Verify Instant Privilege Grant:

**Role:** Switch back to Farmer ( `farmer@agroInk.com` ).

**Visual Check:** Notice the farmer now displays the emerald **Verified Producer** shield badge.

**Action:** Reopen **Post Produce**; the form is now 100% unlocked for trading.

✓ **Expected Outcome:** Secure role-gated onboarding where only KYC-approved producers can initiate commercial trades.

## SCENARIO 2: Spot Produce Listing & 0.50% Escrow Procurement

**Goal:** Test listing creation and verify transparent 0.50% take-rate fee math (0.25% Buyer + 0.25% Farmer).

### Step 2.1 — Farmer Creates a Produce Lot:

**Role:** Farmer ( `farmer@agroInk.com` ).

**Navigation:** Go to **Post Produce** ( `#/farmer-create-listing` ).

**Form Values:**

**Commodity:** `Tomato`

**Variety:** `Hybrid Shivam`

**Grade:** `Grade A Premium`

**Quantity:** `500 kg` | **Unit Price:** `₹42 / kg`

**Location:** `Attur Farmgate Hub, Salem, Tamil Nadu`

**Action:** Click **Publish Listing**.

### Step 2.2 — Buyer Browses & Places Escrow Order:

**Role:** Wholesale Buyer ( `buyer@agroInk.com` ).

**Navigation:** Go to **Marketplace** ( `#/buyer-marketplace` ).

**Action:** Search for `Tomato` or filter by `Tamil Nadu`.

**Action:** Click on the listing card to view details ( `#/buyer-listing-detail` ).

**Action:** Click **Direct Purchase / Buy Now**.

### Step 2.3 — Audit Fee Breakdown in Order Modal:

**Quantity to Buy:** Enter `100 kg`.

**Check Numbers:**

Commodity Subtotal: `100 kg × ₹42 = ₹4,200`

Buyer Platform Escrow Fee (0.25%): `₹10.50`

Total Buyer Deposit: `₹4,210.50`

**Action:** Click **Confirm & Deposit to Escrow**.

✓ **Expected Outcome:** Order status moves to `confirmed` with funds securely held in escrow.

## SCENARIO 3: Pre-Dispatch Assayer Quality Check & Certification

**Goal:** Test buyer quality protection where certified assayers inspect moisture and grade before dispatch.

### Step 3.1 — Buyer Requests Assayer Inspection:

**Role:** Wholesale Buyer ( `buyer@agroInk.com` ).

**Navigation:** Go to **My Orders** ( `#/buyer-orders` ) and click on the newly placed tomato order.

**Action:** Locate the **Quality Inspection Desk** card inside the order summary modal.

**Action:** Click **Inspect Quality (Pre-Dispatch)**.

**Action:** Check parameters ( **Moisture Content** , **Grade Assaying** ) and click **Submit Quality Inspection Request**.

### Step 3.2 — Admin Assayer Completes Physical Assay:

**Role:** Platform Admin ( **admin@agroInk.com** )

**Navigation:** Go to **Dispute & Inspection Desk** ( **#/admin-disputes** ).

**Action:** Find the inspection request for Order #ORD-XXXX and click **Dispatch / Complete Assay**.

**Fill Assay Report:**

**Assay Grade:** **Grade A Premium**

**Moisture Content:** **10.8%** (Optimal)

**Foreign Matter:** **0.3%**

**Action:** Click **Approve & Dispatch Assay Report to Buyer**.

### Step 3.3 — Buyer Views Certified Assay Sheet:

**Role:** Switch back to Buyer ( **buyer@agroInk.com** ).

**Visual Check:** Reopen the order. The inspection button is now replaced by a permanent **Readonly Certified Assay Report** showing all certified parameters.

✓ **Expected Outcome:** Zero quality risk for wholesale buyers through independent pre-dispatch certification.

## 🔧 SCENARIO 4: Real-Time Live Reverse-English Auction Arena

**Goal:** Test live bidding floor with real-time countdown, reserve price checks, and instant win settlement.

### Step 4.1 — Farmer Launches Live Auction Floor:

**Role:** Farmer ( **farmer@agroInk.com** )

**Navigation:** Go to **Create Auction** ( **#/farmer-create-auction** ).

**Parameters:** **Basmati Rice (Pusa 1121)** | **20 Tonnes** | Starting Bid: **₹38 / kg** | Reserve: **₹45 / kg** .

**Action:** Click **Start Live Auction Floor**.

### Step 4.2 — Buyer Enters Arena & Places Competitive Bid:

**Role:** Wholesale Buyer ( **buyer@agroInk.com** )

**Navigation:** Go to **Live Auctions** ( **#/buyer-live-auctions** ) ➡ Click **Enter Auction Arena** ( **#/auction-room** ).

**Action:** Place a bid above reserve price (e.g. **₹46 / kg** ).

**Visual Check:**

Live bid log records the bid timestamp and bidder alias.

Status transitions from **Reserve Not Met** to **Reserve Met ✓** in emerald green.

Real-time timer decrements towards auction close.

✓ **Expected Outcome:** Dynamic, competitive price discovery with automated win conversion into an escrow contract.

## 🔧 SCENARIO 5: Multi-Axle Logistics Manifest & Delivery Confirmation OTP

**Goal:** Test corridor GPS milestone tracking and secure milestone-based escrow fund release.

### Step 5.1 — Transporter Updates Corridor Milestones:

**Role:** Transporter ( **transporter@agroInk.com** )

**Navigation:** Go to **Logistics Corridor Hub** ( **#/transporter-dashboard** ).

**Action:** Select the assigned freight lot.

**Action:** Assign Truck Type: **Multi-Axle 16T Reefer (TN-28-AC-4091)** .

**Action:** Advance transit status: **Picked Up** ➡ **In Transit** ➡ **Arrived at Hub** .

### Step 5.2 — 6-Digit Delivery Confirmation OTP:

**Role:** At the delivery location, Buyer shares the **6-digit delivery OTP**.

**Action:** Transporter enters OTP and clicks **Verify & Complete Delivery**.

Step 5.3 — Verify Automatic Escrow Release:

Role: Platform Admin ( `admin@agroInk.com` ) ➔ Escrow Ledger ( `#/admin-escrow` ).

Visual Check: Order updates to `Completed / Escrow Released to Farmer` with the exact 0.50% fee retained by AgroInk.

✔ Expected Outcome: 100% fraud-proof settlement; farmer receives payout only after verified delivery.

🔑 SCENARIO 6: Anti-Circumvention Privacy Chat & Smart Evasion Shield

Goal: Test that contact evasion tactics (split phone digits, domain keywords, number words) are blocked in real time.

Step 6.1 — Open Direct Privacy Channel:

Role: Wholesale Buyer ( `buyer@agroInk.com` ).

Navigation: Open My Orders ( `#/buyer-orders` ) ➔ Click the 🗨 Privacy Chat drawer button.

Action: Select channel: **Veerappan (Farmer) • Direct Channel**.

Step 6.2 — Test Circumvention Attack Vectors:

Try typing the following messages to test the smart shield:

| TEST INPUT SCENARIO                   | MESSAGE CONTENT TYPED                 | EXPECTED SMART SHIELD MASKING                                                                           |
|---------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------|
| Standard Phone Number                 | Call me at 9840123456 for direct deal | Call me at 98*** ***56 [Protected Phone] for direct deal                                                |
| Email Address                         | Send invoice to buyer.agro@gmail.com  | Send invoice to [Protected Email]                                                                       |
| Domain Fragment                       | Contact me on yahoo or dot com        | Contact me on [Protected Domain] or [Protected Domain]                                                  |
| Split Number Evasion (Sliding Window) | Msg 1: 98755<br>Msg 2: 67890          | Sliding Window Intercepts: Masked to 98*** [Protected Fragment] + Security Warning Bot Alert triggered! |
| Number Words                          | call me at nine eight four zero...    | Words normalized to digits and masked safely.                                                           |

Step 6.3 — Verify Real-Time Bidirectional Sync:

Role: Login as Farmer in another tab/browser ( `farmer@agroInk.com` ).

Visual Check: Both parties see the synchronized conversation thread in real time.

✔ Expected Outcome: Zero PII leakage; traders communicate freely while platform revenue is safeguarded.

🔑 SCENARIO 7: e-NWR Warehouse Deposits & 30-Day Working Capital Financing

Goal: Test electronic Negotiable Warehouse Receipts (e-NWR) and 75% LTV commodity-backed credit underwriting.

Step 7.1 — Farmer Deposits Produce into Cold Storage:

Role: Farmer ( `farmer@agroInk.com` ) ➔ e-NWR Vault ( `#/farmer-inventory` ).

Action: Click + Deposit Produce, select `Salem Agri Cold Hub`, `1000 kg Tomato`, `Chamber B4`.

Visual Check: An electronic title `eNWR-2024-XXXX` is minted.

Step 7.2 — Farmer Applies for 75% LTV Working Capital:

Action: Click Apply for Working Capital on the receipt card.

Action: Request `₹25,000` (Max ceiling: `75% of ₹42,000 = ₹31,500`) for `30 Days`.

Action: Click Submit Working Capital Request.

Step 7.3 — NBFC Financier Approves & Places Collateral Lien:

Role: Financier ( `financier@agroInk.com` ) ➔ Underwriting Desk ( `#/financier-underwriting` ).

Action: Click Review & Underwrite ➔ Click Approve & Disburse Funds.

**Visual Check:** Open **Collateral Vault** ( `#/financier-collateral-vault` ). The receipt is tagged with an active **Lien / Pledge Badge**, locking it from unauthorized withdrawal until repaid.

✓ **Expected Outcome:** Immediate working capital liquidity for farmers without predatory middleman loans.

## SCENARIO 8: Live Escrow Account Gateway API & Realtime Webhooks

**Goal:** Test the virtual nodal Escrow Account engine, simulated bank gateway webhooks, and live balance movements in real time.

### Step 8.1 — Open the Live Escrow Gateway Console:

**Role:** Platform Admin ( `admin@agrolnk.com` )

**Navigation:** Go to **Admin Dashboard** ( `#/admin-dashboard` ) or **Escrow Ledger** ( `#/admin-escrow` ).

**Action:** Click the green  **Live Escrow API** button in the header.

**Visual Check:** The **Live Escrow Account Gateway & Nodal API** modal opens displaying:

**Nodal Partner:** `ICICI Bank Ltd. (Institutional Agri-Nodal Branch)`

**Trustee Partner:** `Vistra / Axis Trustee Services Ltd. (SEBI Reg.)`

**Virtual Nodal Account:** `992010884210982` • IFSC: `ICIC0000104`

**Live Metrics:** Locked in Escrow, Settled Payouts, 0.50% AgroLnk Revenue pool.

### Step 8.2 — Test Inward Escrow Deposit Simulator:

**Navigation:** Switch to the  **Live Interactive Simulator** tab in the modal.

**Action:** Enter Commodity: `Tomato`, Trade Amount: `₹5,000`, Buyer: `Maran S`, Farmer: `Veerappan`.

**Action:** Click  **Trigger Live Inward Deposit**.

**Visual Check:**

Instant toast confirmation: `✓ Captured ₹5,012.50 into Escrow Vault! UTR: CMSICICI...`

Locked vault balance increments immediately.

### Step 8.3 — Test OTP Milestone Payout & Realtime Webhook Bus:

**Action:** Click  **Verify OTP & Disburse to Farmer**.

**Visual Check:** Funds shift from **Locked in Escrow** to **Settled Payouts**, and the 0.50% commission pool increments in real time!

**Action:** Switch to the  **Live Webhook Logs** tab to view the live JSON event bus:

`webhook.escrow.payment_captured`

`webhook.escrow.payout_disbursed`

✓ **Expected Outcome:** Full institutional escrow transparency with real-time auditability and zero settlement leakage.

## 3. Quick Sanity & Production Readiness Checklist

✓ **Zero Build Errors:** `npm run build` compiles 1,948+ modules cleanly with 0 errors.

✓ **Zero SQL Policy Collisions:** Supabase schema uses idempotent drop/create definitions.

✓ **Live Nodal Escrow API:** Real-time ledger, simulated webhooks, and virtual account routing operational.

✓ **Responsive UI:** Tested across Desktop, Tablet, and Mobile viewports.

✓ **Channel Deduplication:** Chat displays 1 entry per business partner regardless of order volume.

✓ **Accurate Revenue Math:** 0.25% Buyer + 0.25% Farmer = 0.50% Total Platform Take-Rate.